BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 281 · Data capture, integrity and ACID transactions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does DELETE differ from DROP TABLE?

Q2. What is a foreign key?

Q3. What does database consistency mean?

Q4. Array scores=[6,2,9,4]; replace element at index2 with element0+element1. Give final array and sum.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Atomicity — All changes or none.

Consistency — Constraints preserved.

Isolation — Concurrent effects safely separated.

Durability — Committed changes persist.

referential integrity — Foreign keys refer to valid rows.

New learning

Understand — Data capture gathers data, while validation, verification and referential integrity address different correctness concerns.

Trace — Atomicity ensures a transaction is all-or-nothing; consistency preserves constraints; isolation addresses concurrent interference; durability keeps committed results.

Explain — Row locking, transactions and appropriate constraints help prevent two concurrent changes from corrupting financial or stock records.

Worked example

A stock purchase needs to reduce Stock and create Order together. Why use a transaction?

1. Predict — Both writes must succeed together or be undone so stock does not change without a valid order.

2. Check the answer — Explain which rule makes this result correct.

We Do · Practise with guidance

Two users attempt to reserve the last seat. Explain a safe database strategy.

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Check and update remaining seats atomically; lock/serialise as required.
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Predict

Only one successful transaction should reserve the final seat, preventing double booking.

Check the answer

Explain which rule makes this result correct.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State the four ACID properties. [4]

Question 2

A transfer starts A100, B50. Move £20 and explain rollback on failure. [6]

Question 3

Two clients try to book one remaining place. Explain how transaction isolation and locking protect integrity. [6]

Exit Ticket · Answer before checking

A failed two-step bank transfer may safely leave the debit saved without the credit.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. DELETE removes row data; DROP removes table definition.

Q2. Reference from one table to another key.

Q3. Constraints remain satisfied by a valid committed transaction.

Q4. scores[2]=6+2=8; array [6,2,8,4]; sum20.

You Do pages · Independent questions

Q1. Atomicity all-or-nothing [1], Consistency constraints [1], Isolation concurrent transactions behave safely [1], Durability committed changes persist [1].

Q2. Initial total150 [1]; A80 [1], B70 [1], use one transaction [1]; if credit fails revert debit [1]; committed total150 [1].

Q3. Read existing capacity [1], competing updates race [1], serialise/lock or atomic conditional update [2], only one succeeds [1], no negative/double booking [1].

Exit Ticket · Model answer

False. Atomicity requires rollback of the partial transfer.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: